12.848

EE25BTECH11002 - Achat Parth Kalpesh

Question:
The tangent plane to the surface $x^2 + y^2 + z = 9$ at the point $(1, 2, 4)$ is

1) $2x+4y+z = 14$ 2) $4x+2y+z = 12$ 3) $x+4y+2z = 17$ 4) $4x+y+2z = 14$

Solution:
The equation of the surface is

$$\mathbf{x}^\top V\mathbf{x} + 2u^\top \mathbf{x} + f = 0 \tag{4.1}$$

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}, \quad f = -9 \tag{4.2}$$

The tangent plane is given as

$$(\mathbf{Vq} + \mathbf{u})^\top \mathbf{x} + \mathbf{u}^\top \mathbf{q} + f = 0 \tag{4.3}$$

where

$$\mathbf{q} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \tag{4.4}$$

$$\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}\right)^\top \mathbf{x} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}^\top \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = 9 \tag{4.5}$$

$$\begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix}^\top \mathbf{x} = 7 \tag{4.6}$$

$$\implies 2x + 4y + z = 14 \tag{4.7}$$

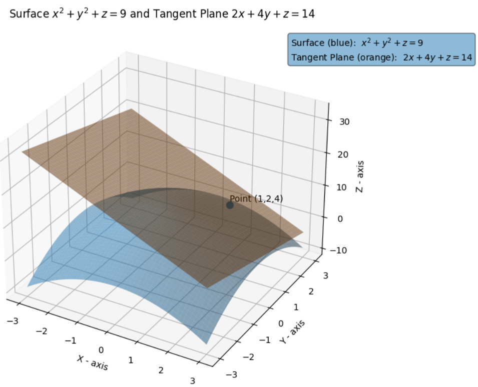
Surface $x^2 + y^2 + z = 9$ and Tangent Plane $2x + 4y + z = 14$
Surface (blue): $x^2 + y^2 + z = 9$
Tangent Plane (orange): $2x + 4y + z = 14$
Point (1,2,4)
X - axis
Y - axis
Z - axis

Fig. 4.1: Graph